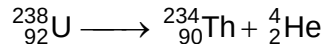


29
L3

Solution: C

The reaction can be represented by



By principle of conservation of momentum,

$$(238 - 4) V_{th} = 4 V_{\alpha}$$

where V_{th} : velocity of the recoiling thorium nucleus and v_{α} : velocity of the α – particle

$$\Rightarrow \frac{V_{\alpha}}{V_{th}} = \frac{234}{4}$$

Ratio of the kinetic energies is

$$\frac{\text{kinetic energy of the } \alpha \text{ – particle}}{\text{kinetic energy of the recoiling thorium nucleus}} = \frac{4V_{\alpha}^2}{234V_{th}^2} = \frac{4}{234} \left(\frac{234}{4} \right)^2 = \frac{234}{4}$$

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Solution: C